ANSWERS:-

1. B. A → C → D, cost 6
2. C. 8
3. C. It needs a usable model of actions and state transitions to plan reliably
4. B. Select the 7-step path if its total cost is lower
5. A. Goal formulation
6. B. Breadth-First Search; finds a shallowest solution and is optimal under equal step costs
7. B. The state may omit information needed to predict actions and outcomes
8. C. Update its representation of the current situation and replan if necessary
9. B. Maintain a set of visited states
10. A. Route X